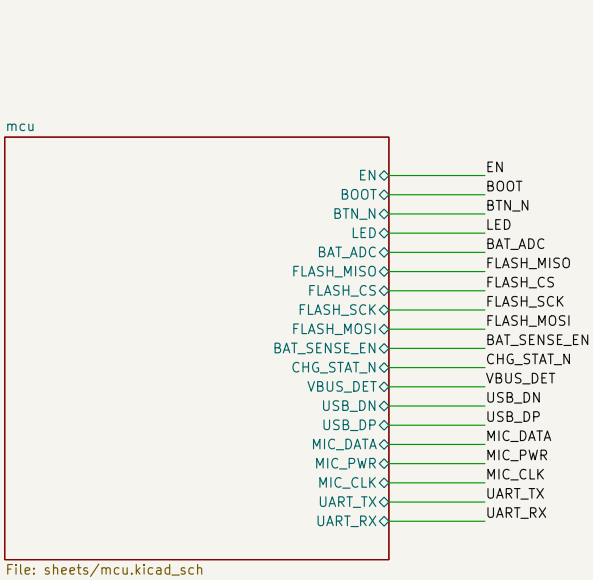
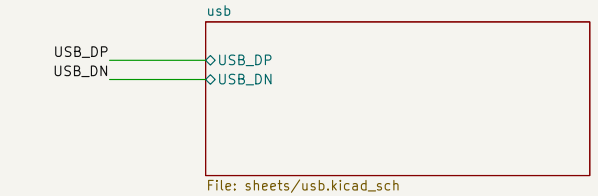
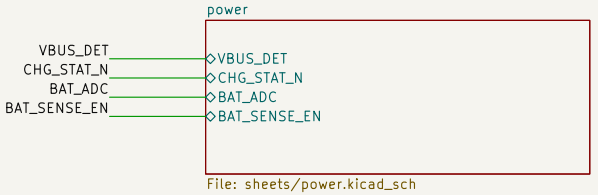
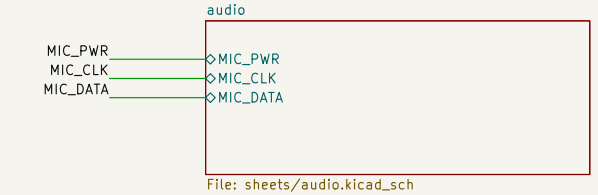
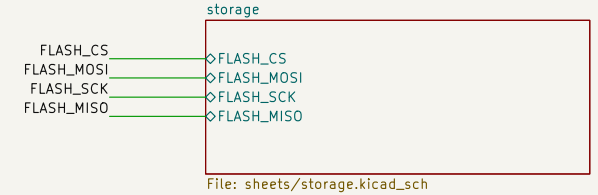
Smart bookmark voice recorder — root sheet. Power nets GND / +3V3 / VBUS / +BATT are global power symbols; signals cross sheets via hierarchical pins joined by the labels here.

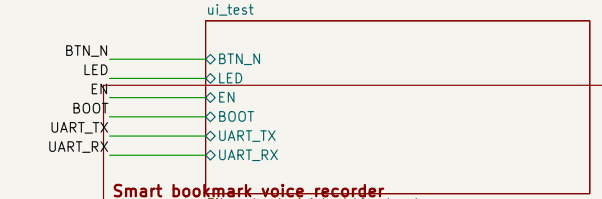












Smart bookmark voice recorder

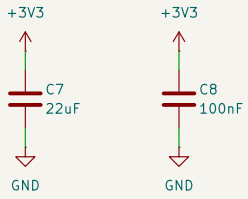
Sheet: /

File: smart-bookmark-esp32s3.kicad_sch

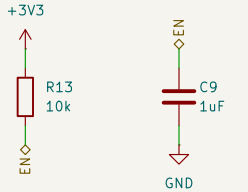
Title: smart-bookmark-esp32s3 — root

Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/7

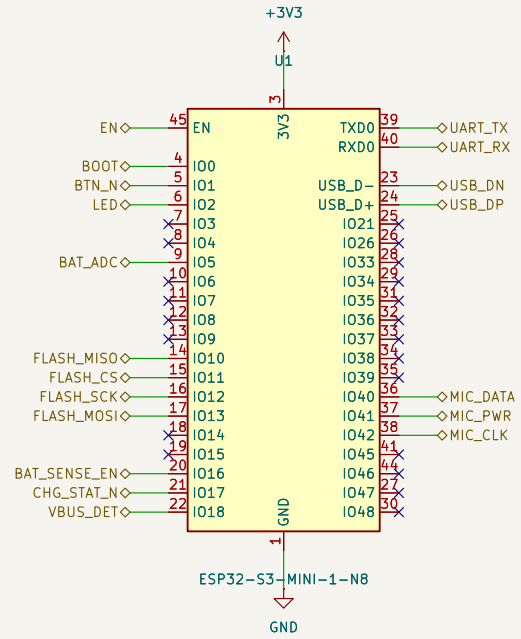
3V3 decoupling at module pin 3



EN power-up delay: 10k / 1 uF



U1 ESP32-S3-MINI-1-N8. Straps: IO0 pull-up (BOOT pad), IO3 float, IO45/IO46 internal pull-down — all left open



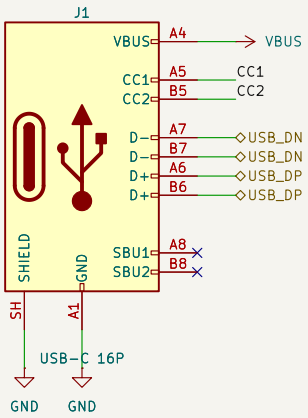
Smart bookmark voice recorder

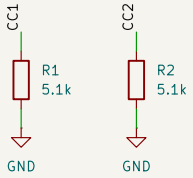
Sheet: /mcu/
File: mcu.kicad_sch

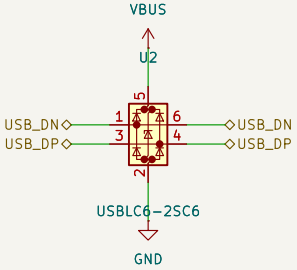
Title: smart-bookmark-esp32s3 — ESP32-S3-MINI-1-N8

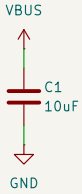
Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/7

J1 USB-C (charge + native USB flashing/logs on GPIO19/20). 5.1k on each CC = sink, 5 V default current







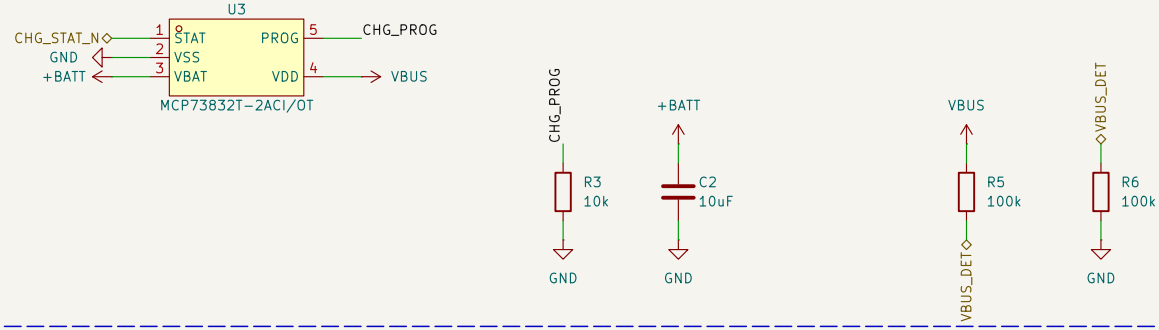




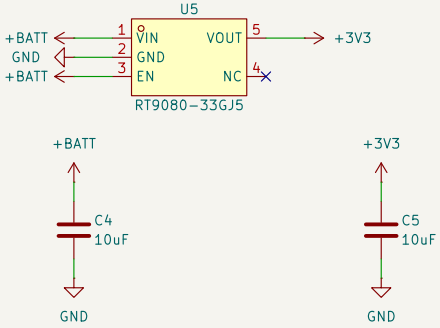
U2 next to J1, D+/D- routed through its pads. C1 is the only bulk on VBUS (USB limit 10 uF).

Smart bookmark voice recorder		
Sheet: /usb/ File: usb.kicad_sch		
Title: smart-bookmark-esp32s3 - USB-C input, CC, ESD		
Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/7

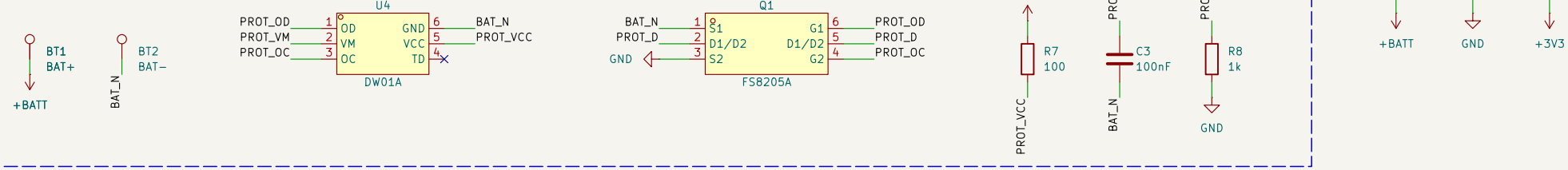
U3 charger: 4.2 V, Ichg = 1000 V / R3 = 100 mA. STAT open-drain, low = charging (MCU pull-up). R5/R6 = USB-present sense



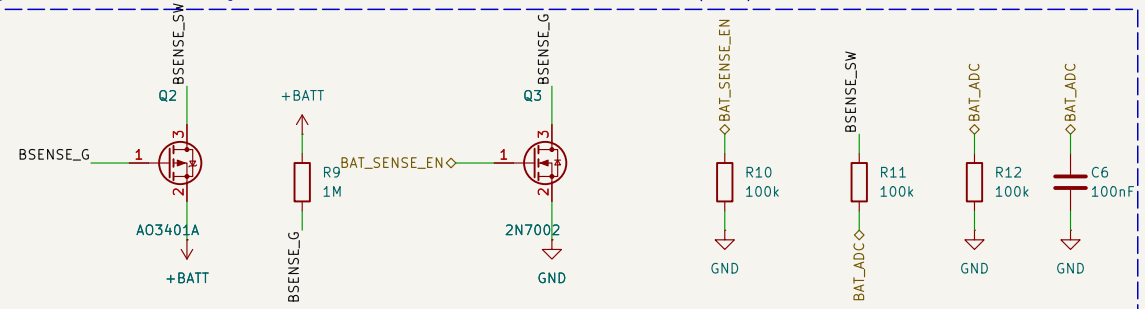
U5 3V3 LDO, 2 uA Iq, always on (EN = +BATT)



Cell protection in the negative leg: BT2 (cell −) = BAT_N -> Q1 -> GND. 2.5 V cut-off, 4.3 V over-charge, short-circuit



Battery sense: BAT_SENSE_EN high -> Q3 on -> Q2 on -> +BATT / 2 on BAT_ADC (ADC1). Off = no divider current



Load sits on the cell (no power path): firmware sleeps when idle

on USB so the charge current can fall to the termination level.

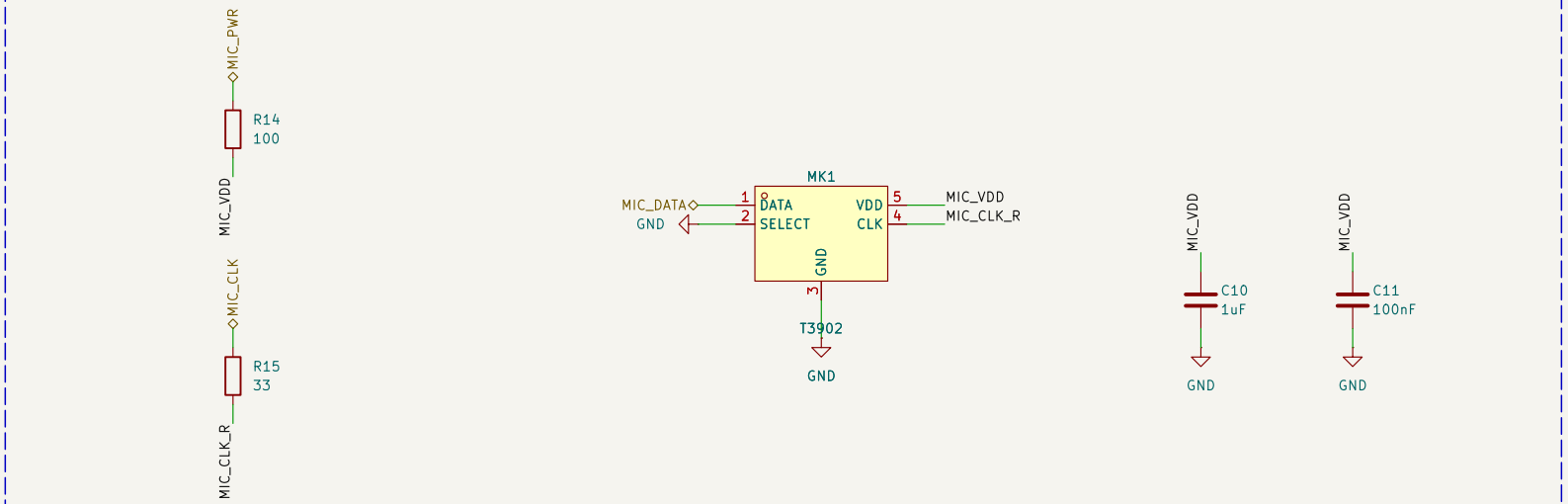
Smart bookmark voice recorder

Sheet: /power/
File: power.kicad_sch

Title: smart-bookmark-esp32s3 — Charger, cell protection, 3V3 LDO, battery

Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/7

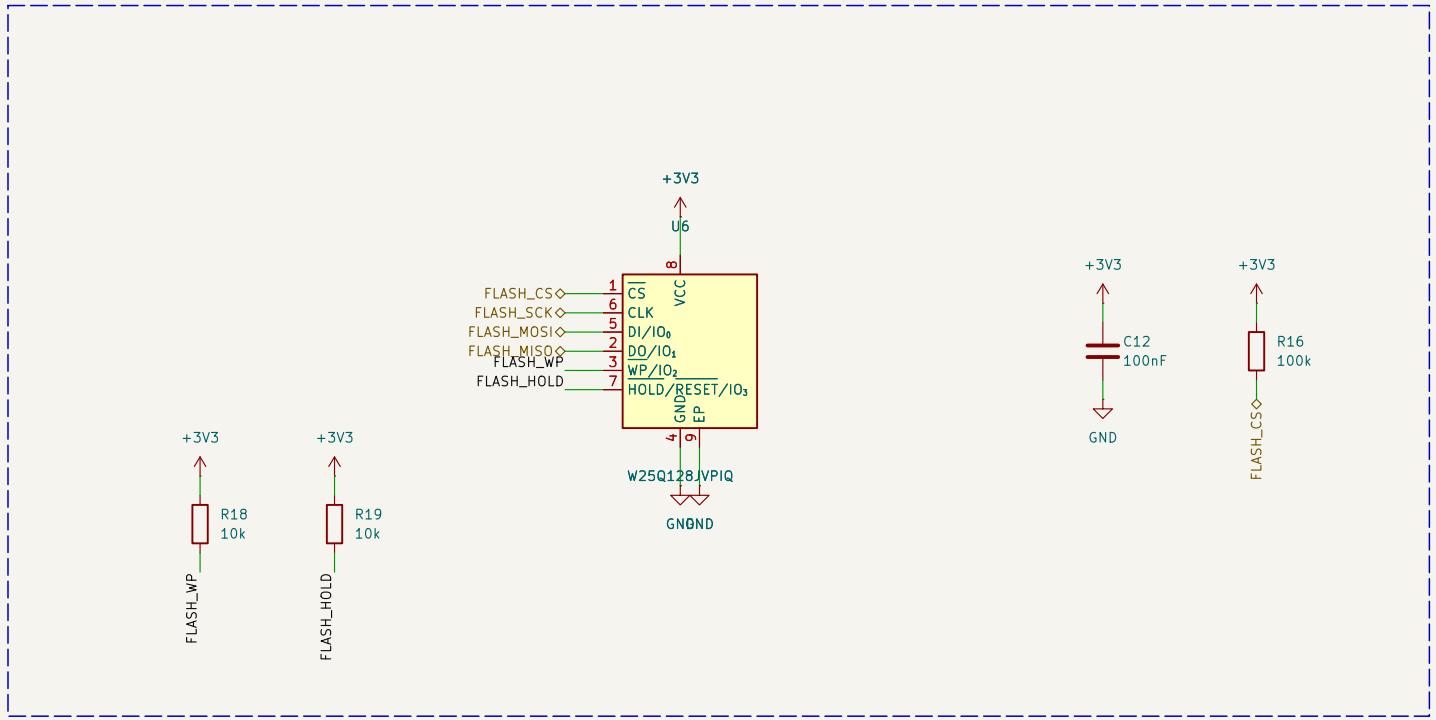
MK1 T3902 bottom-port PDM mic, powered from a GPIO (MIC_PWR) so it draws nothing in deep sleep



SELECT = GND: data valid after the CLK rising edge (right slot). C11 at pin 5. PCB port hole 0.8 mm under the mic.

Smart bookmark voice recorder		
Sheet: /audio/ File: audio.kicad_sch		
Title: smart-bookmark-esp32s3 – PDM MEMS microphone		
Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/7

U6 16 MB NOR on SPI2 for clip storage (about 33 min of audio on top of the module's own flash)



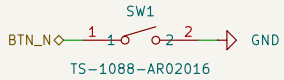
Smart bookmark voice recorder

Sheet: /storage/
File: storage.kicad_sch

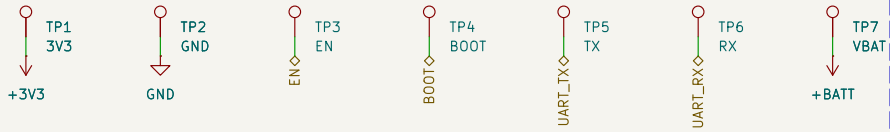
Title: smart-bookmark-esp32s3 – External 16 MB SPI flash for clips

Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 6/7

SW1 record button to GND (internal pull-up, deep-sleep wake on low)

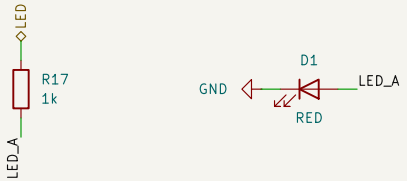


Factory programming / test pads (1.5 mm, bottom side)



Hold BOOT low while pulsing EN low for ROM download mode over UART or USB.

D1 status LED, active high, about 1.3 mA



Smart bookmark voice recorder

Sheet: /ui_test/
File: ui_test.kicad_sch

Title: smart-bookmark-esp32s3 - Button, LED, test pads

Size: A4	Date: 2026-10-05	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 7/7